Small Engines Overhaul · Unit 4 · Level 2 (modified)

Compression



TODAY'S PLAN

- 🗜️ What compression is
- ⭕ What seals the cylinder
- 💨 Causes of low compression
- 📟 The compression test

Review · 5 minutes

Quick Review



List the 4 strokes in order.

What links the piston to the crank?

Which engines use 2-strokes?

Words to Know

Words to Know



Compression

Squeezing the air-fuel mixture into a small space before the spark.



Piston rings

Metal rings around the piston that seal it against the cylinder wall.



Valve seat

The ring the valve presses against to seal the cylinder.



Head gasket

A thin seal between the cylinder head and the block.

Words to Know

More Words to Know



📟 **Compression gauge**: A gauge screwed into the spark plug hole to measure pressure in PSI.

🔢 **PSI**: Pounds per square inch, the unit for pressure.

💨 **Leak**: When pressure escapes past worn rings, valves or a bad gasket.

📉 **Low compression**: Not enough pressure; the engine is hard to start and weak.

Learn

Why Compression Matters



On the compression stroke the mix is squeezed into a small space.

A squeezed mix burns **faster and hotter**, pushing harder.

No compression = hard starting and low power.

Learn

Sealing the Cylinder



Piston rings
seal against the wall



Valves + seats
close tight

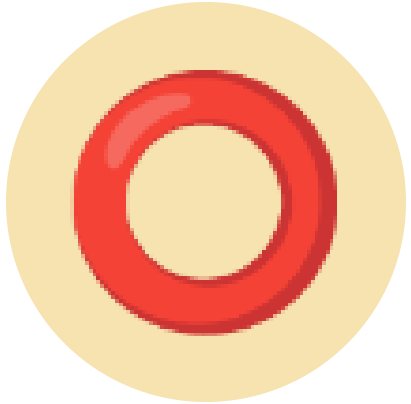


Head gasket
seals head to block



Spark plug
torqued with a washer

🃏 Check 1 of 2



Which part seals the piston to the cylinder wall?

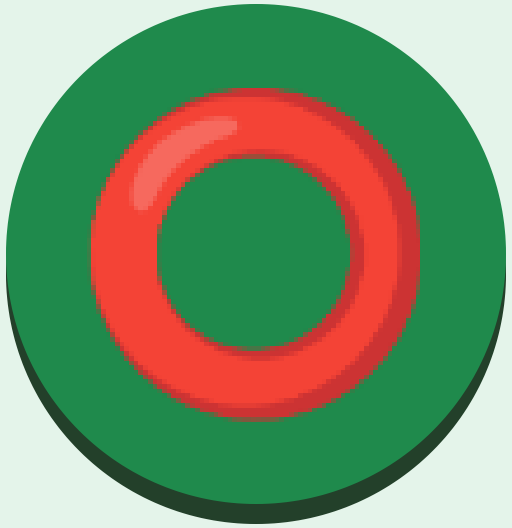
Hold up 1, 2, or 3:

1️⃣ Head gasket

2️⃣ Piston rings

3️⃣ Air filter

✅ Answer · Card 1 of 2



Piston rings

Rings.

Learn

Causes of Low Compression



- Worn or stuck **rings**; scored **cylinder**.
- Burned or carbon-coated **valves**; wrong valve clearance.
- Blown **head gasket** or loose head bolts.

Steps

Compression Test (with adult supervision)



1. Remove the spark plug
2. Screw the gauge into the plug hole
3. Pull the starter rope several times
4. Read and record PSI; compare to spec

🃏 Check 2 of 2



A gauge reads much lower than spec. The engine likely…

Hold up 1, 2, or 3:

1️⃣ Is perfect

2️⃣ Needs paint

3️⃣ Has a leak (rings, valves or gasket)

✅ Answer · Card 2 of 2



Has a leak (rings, valves or gasket)

Low PSI = leak.

Learn

Compression Release



Many small engines have a **compression release** to make pulling easier.

So small-engine gauge readings can look low.

Always use the **manual's spec** for that engine.

Wrap-Up

Recap



Compression
squeeze = power



Seal
rings, valves, gasket



Low
leaks



Test
gauge in plug hole

Wrap-Up

Exit Ticket



Name 3 parts that seal the cylinder.

Name 2 causes of low compression.

Where does the gauge go?